

Diabetes Prediction from Clinical Measurements

Comparing five machine learning algorithms on the Pima Indians dataset

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CBIO313 - Data Mining & Machine Learning

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The problem

Why this matters

- Type 2 diabetes develops **silently**. By the time symptoms appear, damage to the kidneys, retina and nerves is often already underway.
- Screening programmes exist for this reason - but a full diagnostic workup is expensive and cannot be given to everyone.
- **The goal:** a cheap triage step that flags who most needs that workup, using measurements already taken at a routine visit.

The two errors are not equally costly

False positive -> a healthy patient goes for one extra blood test. Inconvenient.

False negative -> an undiagnosed diabetic is sent home reassured. Serious.

Objectives

- Clean a dataset where missing values were silently recorded as zeros
- Explore how each measurement relates to diabetes status
- Train and compare **five** algorithms from the course
- Evaluate with metrics suited to imbalanced medical screening - not accuracy alone
- Tune the best model and choose a threshold appropriate for screening
- Explain *why* the model predicts what it predicts

The dataset

Pima Indians Diabetes Database - UCI / NIDDK

768

patients

8

features

35%

have diabetes

21+

women, Pima heritage

Eight measurements: pregnancies, plasma glucose, blood pressure, skin-fold thickness, serum insulin, BMI, family-history score, age.

Target: Outcome - 1 = diabetes, 0 = no diabetes. 500 negative / 268 positive.

All patients are women of Pima heritage aged 21+ near Phoenix, Arizona - a narrow population, revisited in the limitations.

Finding 1: the zeros are not zeros

The most important preprocessing step

Five columns report a minimum of **0** - glucose, blood pressure, skin thickness, insulin, BMI. None is biologically possible in a living patient.

These are missing measurements that were never marked as missing. Left alone, a model reads *insulin* = 0 as a real, extremely low reading - the opposite of the truth.

Column	Zeros	% of rows
Insulin	374	48.7%
SkinThickness	227	29.6%
BloodPressure	35	4.6%
BMI	11	1.4%
Glucose	5	0.7%

Dropping incomplete rows would throw away nearly half the dataset.

Instead: mark as NaN, then impute the median inside the model pipeline.

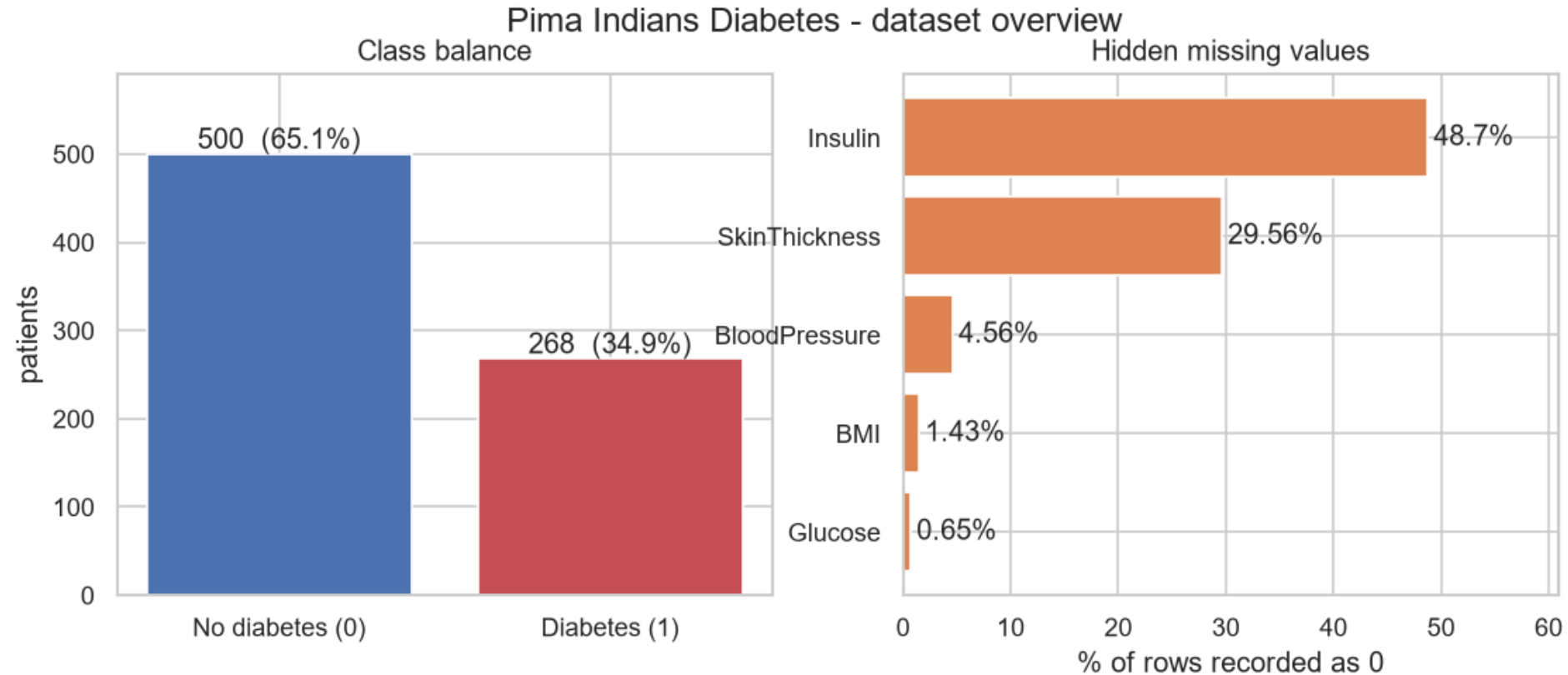
Pipeline: impute (median) → scale (StandardScaler) → classify. One estimator, identical for all five algorithms.

Three places this project could have quietly cheated

- **Imputation is inside the pipeline.** Taking the median over all 768 rows and then splitting would leak test data into training and inflate every score.
- **The winner was chosen on cross-validated training score**, never on test performance. Picking by test score is model selection on the test set.
- **The threshold was chosen from out-of-fold training predictions.** The test set was used exactly once, at the very end.

Exploratory data analysis

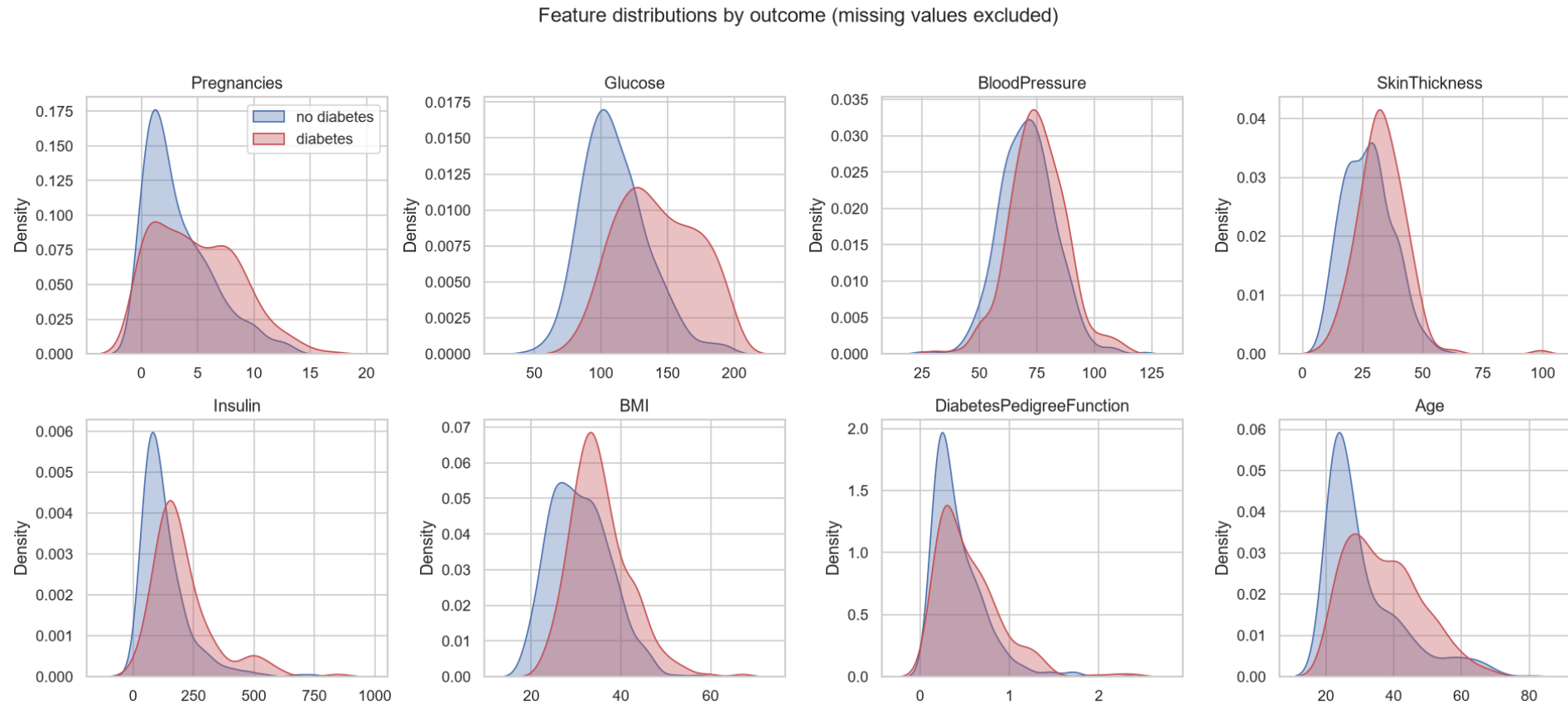
Class balance and missing data



A model that always says "no diabetes" scores 65% accuracy and catches zero diabetics.

That is the benchmark every accuracy figure must be read against.

Which measurements separate the groups?



Glucose separates the two groups most clearly. BMI and Age shift moderately. BloodPressure and SkinThickness overlap almost entirely - they will contribute little.

Five algorithms

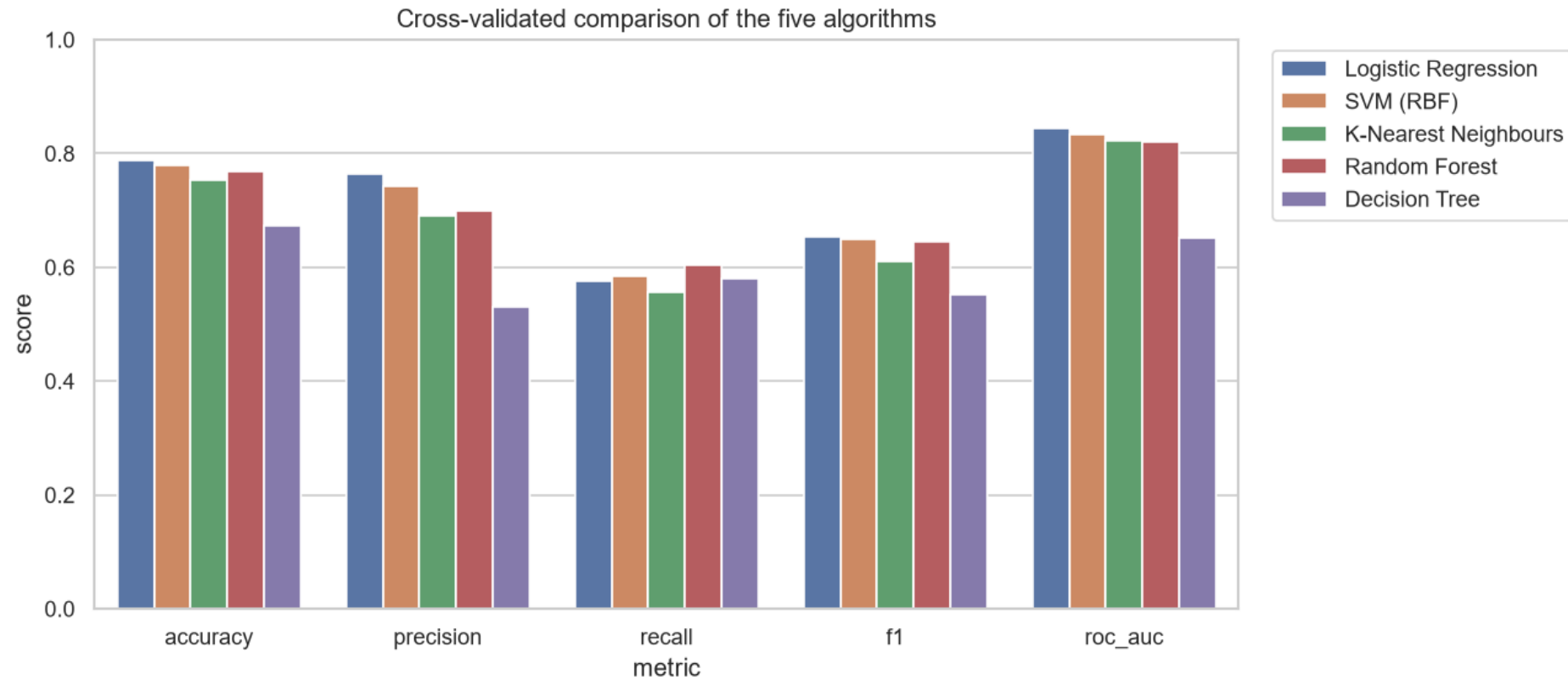
Each in the identical impute - scale - classify pipeline

Algorithm	Why include it	CV ROC-AUC
Logistic Regression	Linear baseline, interpretable coefficients	0.843
SVM (RBF)	Maximum margin with a non-linear kernel	0.833
K-Nearest Neighbours	Non-parametric, purely local decision rule	0.822
Random Forest	Bagged ensemble of 300 trees	0.821
Decision Tree	Interpretable rules, reference for the ensemble	0.651

Stratified 5-fold cross-validation on the training set only - the test set stays sealed.

Model comparison

Cross-validated on the training set

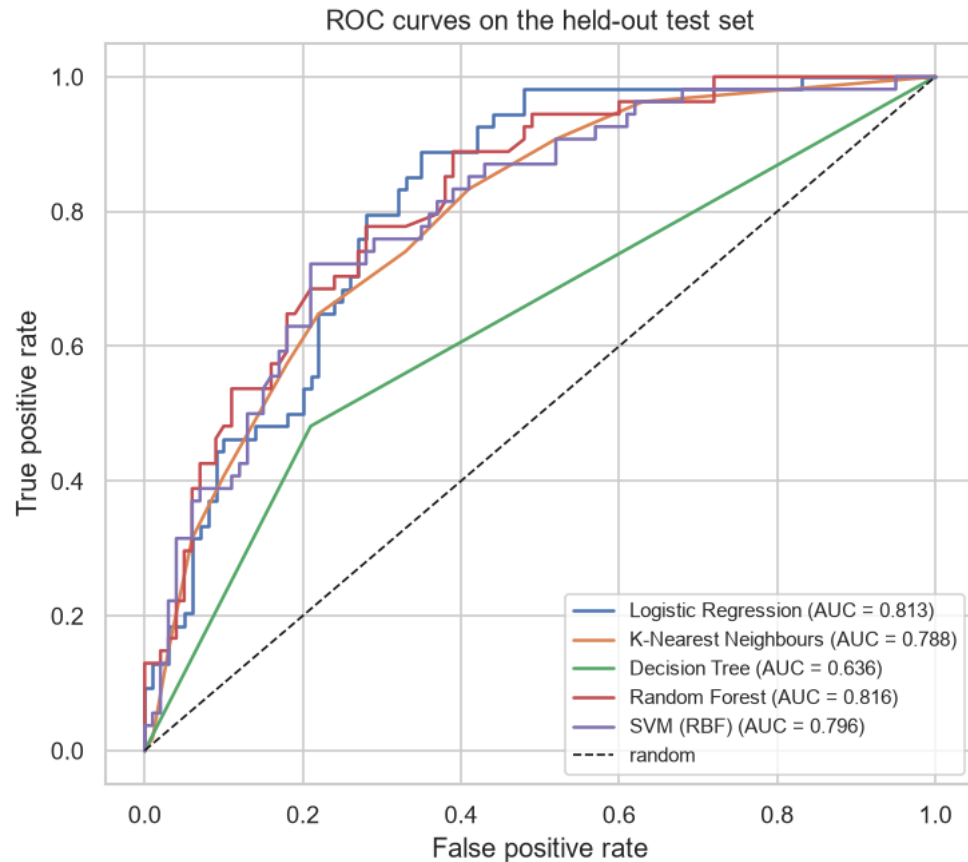


Notice: recall is far below accuracy for every single model.

At the default cut-off these classifiers catch only about 55-60% of diabetics.

Held-out test set

ROC curves - the test set's one and only use



Model	Acc.	Recall	ROC-AUC
Random Forest	0.740	0.537	0.816
Logistic Regression	0.708	0.500	0.813
SVM (RBF)	0.740	0.556	0.796
K-Nearest Neighbours	0.734	0.574	0.788
Decision Tree	0.682	0.481	0.636

Hyperparameter tuning

GridSearchCV, scored on ROC-AUC

Grids are scored on **ROC-AUC, not recall**. Optimising directly on recall rewards models that just predict "diabetic" more often - in the limit, flagging everyone scores perfect recall. ROC-AUC ranks models independently of the threshold.

Model	Best parameters	CV ROC-AUC	Test ROC-AUC
Logistic Regression	C=0.1, class_weight=balanced	0.8450	0.8104
Random Forest	class_weight=None, max_depth=5, min_samples_leaf=3, n_estimators=4000.8388	0.8388	0.8119
SVM (RBF)	C=1, class_weight=balanced, gamma=0.01	0.8493	0.8096

Selected: SVM (RBF) - CV ROC-AUC 0.849, test ROC-AUC 0.810

Finding 2: the threshold is the real result

Where the marks are

predict() flags a patient at probability 0.50. That cut-off maximises accuracy - the wrong target for screening. We chose the threshold from out-of-fold training predictions to reach 80% recall.

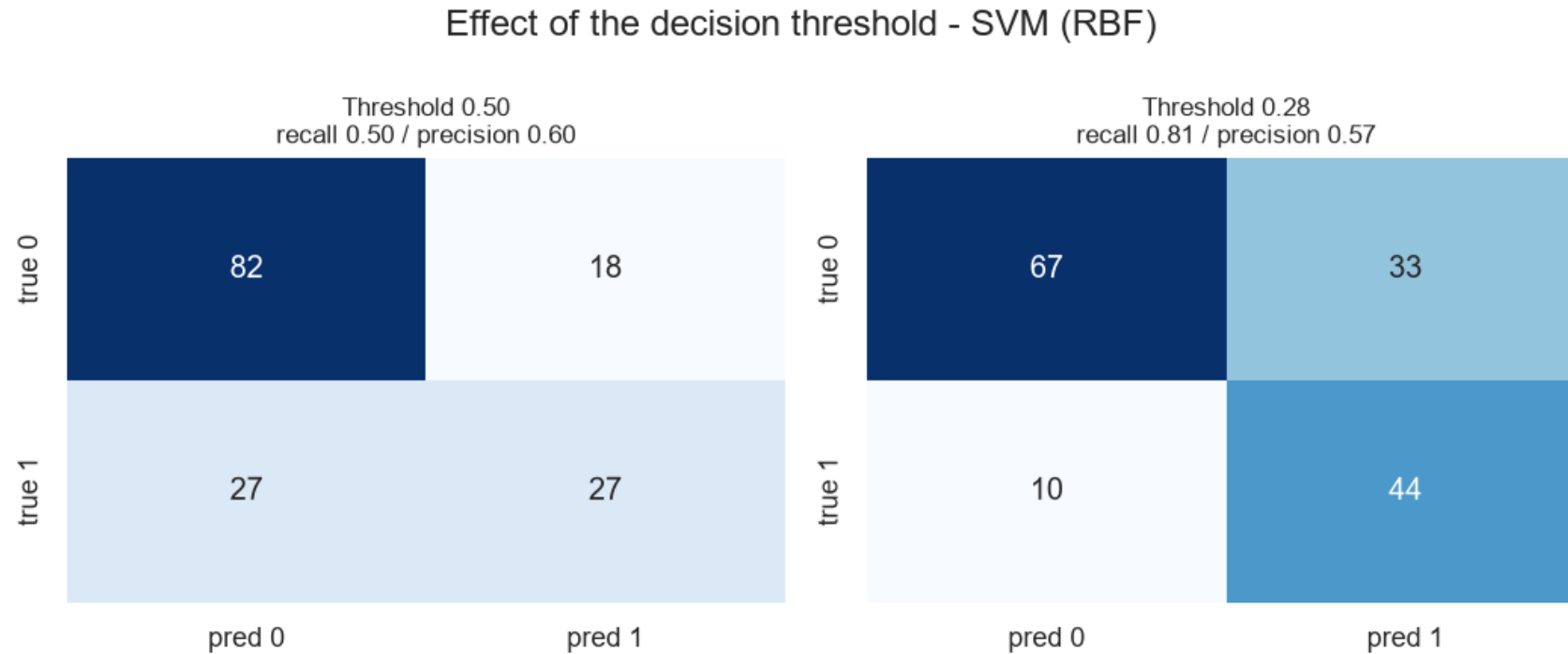
Operating point	Accuracy	Precision	Recall	Missed diabetics
Default 0.50	0.708	0.600	0.500	27 of 54
Chosen 0.285	0.721	0.571	0.815	10 of 54

Accuracy moved 1.3 points. Recall moved 31 points.

Missed diabetics fell from 27 to 10, for a 2.9-point precision cost.

Accuracy was simply not measuring the thing we care about.

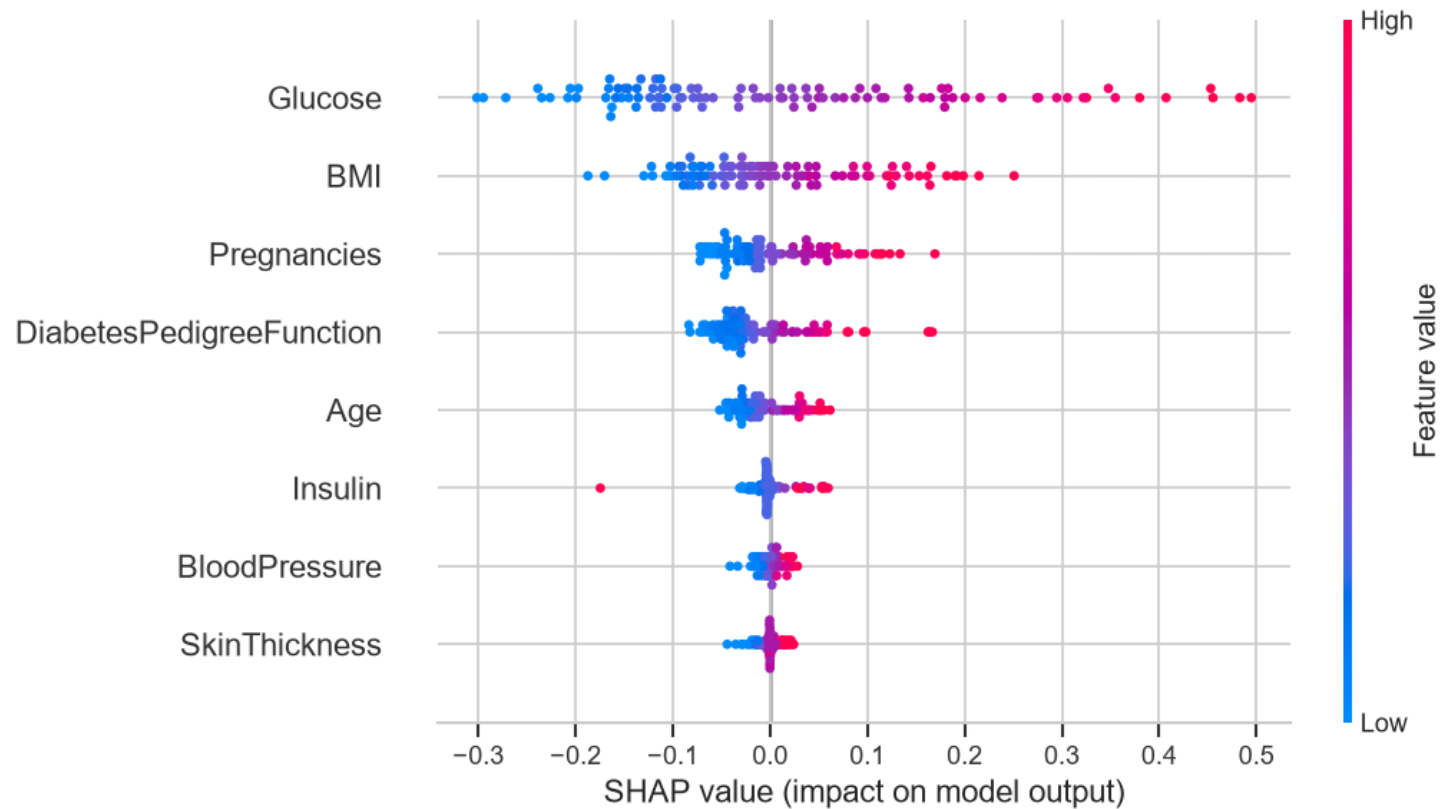
The same model, two operating points



Bottom-left cell = diabetics the model sent home. That is the number to watch.

Explainability

Permutation importance and SHAP agree



- Each dot is one patient
- Right = pushed towards diabetes
- Red = high value, blue = low

Glucose, then BMI

High glucose pushes towards diabetes; low glucose pulls back. The direction is clinically correct - a model with the right accuracy but an inverted relationship would be dangerous, and this plot would expose it.

Live demonstration

Bonus: Streamlit prediction app

- Enter eight measurements on the sidebar
- Get a predicted probability and a screening decision at the tuned threshold
- See how the patient compares to the population
- See a **per-patient SHAP explanation** of that specific prediction

Demo

```
streamlit run src/app.py
```

Default patient: 23.5% probability - not flagged.

Raise glucose to 190: 78.6% - flagged for confirmatory testing.

Limitations

What this model cannot do

- **Narrow population.** 768 patients, all women of Pima heritage aged 21+. Nothing transfers to men, other ethnicities, or younger patients.
- **Heavy imputation.** 49% of insulin and 30% of skin-fold values were filled in - and both rank near the bottom of the importance plots.
- **Missingness is probably not random.** Median imputation assumes it is.
- **No external validation.** One 154-patient test split, so the interval around a recall of 0.81 is wide.
- **Information ceiling.** Eight coarse measurements carry limited signal. Better results need better features (HbA1c, fasting glucose), not fancier models.

This is a course project, not a medical device.

Conclusion

0.810

test ROC-AUC

81%

of diabetics caught

10 of 54

missed, down from 27

5

algorithms compared

- The gains came from **preprocessing** and from **matching the operating point to the clinical cost of each error** - not from the choice of algorithm.
- Four of the five models landed within a few points of each other. A well-regularised linear model was competitive with everything more elaborate.
- As a triage aid flagging ~80% of diabetics for confirmatory testing, the model is plausible and its behaviour is explainable.

Thank you

Questions?

github.com/ssameh12/diabetes-prediction-cbio313

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